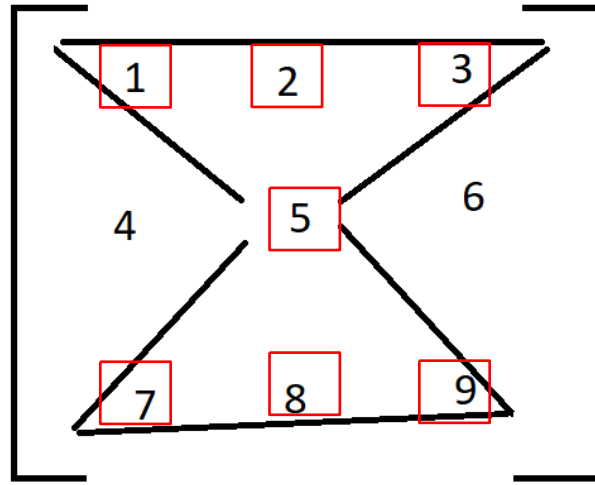




# Maximum sum of hour glass in matrix

The concept of finding the maximum sum of an hourglass in a matrix involves traversing the elements of the matrix and identifying hourglass patterns. An hourglass pattern refers to a specific arrangement of elements in the matrix that resembles the shape of an hourglass. Here's an example of an hourglass pattern:





$$\text{Sum} = 1+2+3+5+7+8+9 = 35$$

Now What is Mean By Maximum Sum of Hour Glass?

3x3 has only 1 Hour glass but,

4x4 has 4 hour glass  
for eg:

|   |   |   |   |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 1 | 2 | 3 |
| 4 | 5 | 6 | 7 |

we have to find max sum of this 4 possibilities

$$\text{sum1} = 1+2+3+6+9+1+2 = 24$$

$$\text{sum2} = 2+3+4+7+1+2+3 = 22$$

$$\text{sum3} = 5+6+7+1+4+5+6 = 34$$

$$\text{sum4} = 6+7+8+2+5+6+7 = 41$$

so, Return 41 the maximum sum in the given 4x4 matrix

how to find number of possibilities?

we know that:

$$(R-2) (C-2)$$

$$\text{can get } 4 \times 4 = (4-2)(4-2) = 4$$

```
1. import java.util.Scanner;
2. public class Hourglass {
3.     static int findMaxSum(int [][]mat,int R,int C)
4.     {
5.         int max_sum=0;
6.         int sum;
7.
8.         if (R < 3 || C < 3){
9.             System.out.println("Not possible");
10.            System.exit(0);
11.        }
12.        for (int i = 0; i < R - 2; i++)
13.        {
14.            for (int j = 0; j < C - 2; j++)
15.            {
16.                sum = (mat[i][j] + mat[i][j + 1] +
17.                    mat[i][j + 2]) + (mat[i + 1][j + 1]) +
18.                    (mat[i + 2][j] + mat[i + 2][j + 1] +
19.                    mat[i + 2][j + 2]);
20.                max_sum = Math.max(max_sum, sum);
21.            }
22.        }
23.        return max_sum;
24.    }
25. }
```

```
1. static public void main (String[] args)
2.     {
3.         Scanner sc = new Scanner(System.in);
4.         int R = sc.nextInt();
5.         int C= sc.nextInt();
6.         int [][]mat = new int[R][C];
7.         for (int i = 0; i < R; i++) {
8.             for (int j = 0; j < C; j++) {
9.                 mat[i][j]= sc.nextInt();
10.            }
11.        }
12.        int res = findMaxSum(mat,R,C);
13.        System.out.println("Maximum sum of hour glass = "+ res);
14.    }
15.
16.}
```



# THANK YOU